

## Class 29: The Characteristic Polynomial

**Definition:** If  $A$  is a square matrix, we define an characteristic polynomial for  $A$  to be

The characteristic equation is

The eigenvalues for  $A$  are the zeros for the characteristic equation.

**Practice:** Find the characteristic polynomial for  $A = \begin{bmatrix} 1 & 5 \\ 2 & 4 \end{bmatrix}$

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 5 \\ 2 & 4-\lambda \end{vmatrix} =$$

Set this equal to zero and solve for the two eigenvalues.

Find an eigenvector for  $\lambda = 6$  by finding a basis for the null space of the matrix  $A - 6I$

Find an eigenvector for  $\lambda = -1$  by finding a basis for the null space of the matrix  $A + I$

These two vectors are linearly independent so they form a basis for  $\mathbf{R}^2$ .

The vector  $\begin{bmatrix} -3 \\ 11 \end{bmatrix} = 7 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} -5 \\ 2 \end{bmatrix}$

Calculate  $A^5 \begin{bmatrix} -3 \\ 11 \end{bmatrix} =$

Let  $P = \begin{bmatrix} 1 & -5 \\ 1 & 2 \end{bmatrix}$ . Then  $P^{-1}AP = \begin{bmatrix} 6 & 0 \\ 0 & -1 \end{bmatrix}$

**Definition:** Matrix  $A$  is similar to matrix  $B$  if

If  $A$  is similar to  $B$ , then it will be true that  $B$  is also similar to  $A$ . We say that  $A$  and  $B$  are similar to each other.

Notice in the example that  $A$  and  $B$  had the same eigenvalues. That is always true for similar matrices.

**Theorem:** If  $A$  and  $B$  are similar, then  $A$  and  $B$  have the same characteristic polynomial and hence the same eigenvalues.

**Proof:**

**Big theorem:** If  $\mathbf{v}_1, \dots, \mathbf{v}_r$  are eigenvectors that correspond to distinct eigenvalues for matrix  $A$ , then the set  $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$  is linearly independent.

**Proof by contradiction:**

